

CPSC 250

Advanced Linear Regression Concepts

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1 Motivation

In earlier lectures, we considered simple linear regression involving one independent variable.

Real-world data analysis is often more complicated.

In this lecture, we discuss:

- multiple regression variables,
- parameter interpretation,
- correlations between variables,
- matrix representations,
- and data structures for regression.

2 Complex Regression Problems

Suppose we want to predict:

$$y$$

using several variables:

$$x_1, x_2, x_3, x_4$$

For example:

- y : housing price,
- x_1 : square footage,
- x_2 : zip code,
- x_3 : number of bedrooms,
- x_4 : number of bathrooms.

3 The Regression Equation

We seek an equation of the form:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

This is linear in the parameters.

4 Why This Is Difficult

Visualizing this problem directly would require a very high-dimensional plot.

For example:

- one dependent variable,
- four independent variables.

This is impossible to visualize completely.

Therefore, regression becomes both:

- a computational problem,
- and an interpretation problem.

5 Parameter Uncertainties

In practice, we determine:

$$\beta_0 \pm \Delta\beta_0$$

$$\beta_1 \pm \Delta\beta_1$$

and so on.

A parameter consistent with zero may indicate that the corresponding variable is not important.

For example:

If the coefficient for number of bedrooms is consistent with zero, housing prices may not strongly depend on that variable.

6 Exploratory Data Analysis

Before fitting, we should first examine the data.

For example:

$$y \text{ vs. } x_1$$

$$y \text{ vs. } x_2$$

and so on.

This helps identify:

- trends,
- bad data,
- outliers,
- nonlinear behavior.

7 Correlations Between Variables

Independent variables may themselves be correlated.

For example:

$$x_2 \text{ vs. } x_3$$

might show a strong trend.

This can produce misleading results.

8 Correlation vs. Causation

An important warning:

Correlation does not imply causation.

Two variables may move together without one causing the other.

Regression analysis must therefore be interpreted carefully.

9 Data Structures for Regression

There are two common choices:

1. NumPy arrays,
2. pandas DataFrames.

10 Matrix Interpretation

Suppose:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$$

for many data points.

We can write:

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{21}$$

$$y_2 = \beta_0 + \beta_1 x_{12} + \beta_2 x_{22}$$

and so on.

11 Matrix Form

These equations can be rewritten as:

$$\mathbf{y} = X\beta$$

where:

- $\mathbf{y}$ is the vector of dependent values,

- X is the design matrix,
- β is the parameter vector.

12 Solving for the Regression Parameters

The matrix equation

$$\mathbf{y} = X\beta$$

looks deceptively similar to an ordinary algebraic equation. Unfortunately, we cannot simply "divide by X " because X is generally *not* a square matrix.

Suppose there are N data points and M fitting parameters. Then

$$X$$

has dimensions

$$N \times M.$$

For example, if we fit a straight line,

$$y = \beta_0 + \beta_1 x,$$

to one hundred data points, then

$$X$$

has dimensions

$$100 \times 2.$$

Since X is not square, the inverse

$$X^{-1}$$

does not exist.

The key idea is therefore to transform the problem into one involving a *square* matrix.

We do this by multiplying both sides of the equation by the transpose of X :

$$X^T \mathbf{y} = X^T X \beta.$$

Now examine the dimensions:

$$X^T \rightarrow M \times N$$

and

$$X \rightarrow N \times M.$$

Therefore,

$$X^T X$$

has dimensions

$$(M \times N)(N \times M) = M \times M.$$

This is now a square matrix.

Provided the columns of X are linearly independent, the matrix $X^T X$ is invertible, allowing us to solve for the unknown parameter vector:

$$\boxed{\boldsymbol{\beta} = (X^T X)^{-1} X^T \mathbf{y}}$$

This equation is known as the *normal equation* for linear least-squares regression.

The Big Picture

One of the most important ideas in this course is that we have transformed a data-analysis problem into a linear algebra problem.

Originally, our goal was to determine the unknown regression parameters $\beta_0, \beta_1, \dots$ that best describe a set of measured data.

By constructing the design matrix X , we reformulated the problem as

$$\mathbf{y} = X\boldsymbol{\beta},$$

and ultimately reduced the entire regression problem to computing the inverse of the matrix

$$X^T X.$$

Thus, from a computational point of view,

Linear regression is fundamentally a matrix inversion problem.

Modern software packages such as `statsmodels` and `scikit-learn` hide these details from the user, but this is the underlying mathematics that they are performing behind the scenes.

In principle:

$$\boldsymbol{\beta} = (X^T X)^{-1} X^T \mathbf{y}$$

This is the mathematical foundation of many regression algorithms.

13 Important Practical Point

For most Python regression tools:

- X -data should be a two-dimensional matrix,
- y -data should be a one-dimensional array.

14 Simple NumPy Example

```
import numpy as np

xi = [1, 2, 3, 4, 5]

yi = [1.1, 2.1, 2.9, 3.9, 5.1]

X = np.array(xi).reshape((-1,1))

y = np.array(yi)
```

15 Why Reshape?

The line:

```
reshape((-1,1))
```

converts the array into a column matrix.

This is required because regression libraries expect:

$$N \times M$$

data structures for independent variables.

16 Data Cleaning

Before fitting, we should:

- plot the data,
- look for missing points,
- identify outliers,
- think about appropriate models.

17 Putting It All Together

We now have all of the pieces needed to perform multiple linear regression on a real dataset.

Suppose a company records the following information for many advertising campaigns:

- television advertising budget,
- radio advertising budget,
- newspaper advertising budget,
- resulting product sales.

Our goal is to determine how strongly each advertising medium influences sales. Using the notation developed above, the regression model becomes

$$\text{Sales} = \beta_0 + \beta_1(\text{TV}) + \beta_2(\text{Radio}) + \beta_3(\text{Newspaper}).$$

Each row of the design matrix corresponds to one advertising campaign, while each column corresponds to one explanatory variable (plus the column of ones used to determine the intercept).

Fortunately, Python libraries such as `statsmodels` perform the matrix computations automatically. The following example illustrates how this is done in practice.

18 Key Takeaways

- Real regression problems often involve many variables.
- Regression coefficients have uncertainties.
- Correlations between variables can complicate interpretation.
- Regression is fundamentally a linear-algebra problem.
- Most regression libraries require two-dimensional X -data.
- Exploratory plotting is extremely important before fitting.

11 Matrix Formulation

The system of normal equations in Section 10 is the matrix form of the OLS equations. To see how this works, we first express the regression model in matrix form:

FROM THE INDIVIDUAL MODEL (Section 10) TO MATRIX FORM ($y = X\beta + \epsilon$)

(A) Individual Model (with intercept)

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

for $i = 1, \dots, n$

Example with $p = 2$ predictors (x_1, x_2):

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \epsilon_n \end{aligned}$$

(B) Stack the n equations

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} \\ 1 & x_{21} & x_{22} \\ \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$\mathbf{y}$
 $n \times 1$
(response)

$\mathbf{X}$
 $n \times (p+1)$
(design matrix)

β
 $(p+1) \times 1$
(coefficients)

ϵ
 $n \times 1$
(errors)

First column of 1's for the intercept!

MATRIX FORM ($y = X\beta + \epsilon$)

$\mathbf{y}$: n -vector of responses (one for each observation)
 $\mathbf{X}$: $n \times (p+1)$ design matrix (one row per observation)
 β : $(p+1)$ -vector of coefficients (including β_0)
 ϵ : n -vector of errors

$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$

Structure of $\mathbf{X}$ (the design matrix)

- Column 1 is all 1's (for the intercept β_0)
- Columns 2 to $p+1$ are the predictors $x_1, x_2, \dots, x_p$

DIMENSIONS

$\mathbf{y}$	$n \times 1$
$\mathbf{X}$	$n \times (p+1)$
β	$(p+1) \times 1$
ϵ	$n \times 1$

The product $\mathbf{X}\beta$ is $n \times 1$, so it can be added to ϵ ($n \times 1$) to give $\mathbf{y}$ ($n \times 1$).

WHAT DOES $\mathbf{X}\beta$ LOOK LIKE?

Compute $\mathbf{X}\beta$:

$$\begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix} = \beta_0 + \beta_1 x_{11} + \dots + \beta_p x_{1p}$$

So, $\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \mathbf{X}\beta = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \dots + \beta_p x_{1p} \\ \beta_0 + \beta_1 x_{21} + \dots + \beta_p x_{2p} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \dots + \beta_p x_{np} \end{bmatrix}$

These are the fitted (predicted) values $\hat{y}_i$ from the regression model.

KEY TAKEAWAYS

- Writing the model in matrix form lets us use linear algebra to derive the OLS estimator.
- The column of 1's in $\mathbf{X}$ is what allows us to estimate the intercept β_0 .
- The normal equations are $(\mathbf{X}^T \mathbf{X})\beta = \mathbf{X}^T \mathbf{y}$, which we solved in Section 10.

THE NORMAL EQUATIONS

$$(\mathbf{X}^T \mathbf{X})\beta = \mathbf{X}^T \mathbf{y}$$

(from Section 10)

Figure 1: Moving from the individual regression equations to the matrix equation $\mathbf{y} = \mathbf{X}\beta + \epsilon$. Notice that the first column of the design matrix $\mathbf{X}$ consists entirely of ones. This column corresponds to the intercept coefficient β_0 , which explains why packages such as `statsmodels` require us to explicitly add a constant column using `sm.add_constant(X)`.